

time to get familiar development tools ported to ARM. Some of the best feedback we have had from developers and testers is that they didn't even realise they were running on ARM hardware," explains Power.

Intel is also talking of a 6W processor for servers now, so how would ARM fare compared to that? While Intel's move is a clear acknowledgement of the need for low-power servers, there is still a lot to be done before Intel's 6W processor can compare with a 5W server. Viridis is a 5W server, which includes all the components required for operation in a data centre (power measured at the wall, including motherboard, network, cooling fans, memory, disk, etc). "When you add all such factors to Intel's processor, it will be significantly higher than 5W. Intel will undoubtedly introduce a SoC product down the line, but at that point we expect to have Viridis v2 (or possibly v3) ready for the market, which will have further performance/watt improvements," says Power.

Others also appear to be in favour of ARM at present. Dell announced the Copper platform in May 2012. This is a platform based on an ARM-powered SoC from Marvell. Mitac in Taiwan also announced and demonstrated a Marvell-powered server at Computex in June 2012. HP has shipped a number of Calxeda-based development platforms (called Redstone) to enable companies to start porting and evaluating the benefit of ARM-based server technology for their applications. A US company called Penguin Computing, focused on data analytics platforms, is also shipping a Calxeda-powered platform. Dell has announced a programme called Iron, using a 64-bit server SoC device from Applied Micro, which is likely to ship later this year.

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The same group has a newly born 'National Software Platform' initiative, which was formed last summer. The objectives of this initiative were declared to be too broad—ranging from the creation of a national Linux distribution (another one, in addition to existing MCBC and AltLinux?) to setting up the policy on how to use (FOSS) within the government sector. It's hard to believe this national platform will be effective, because the participants are the same RASPO mentioned earlier, plus a number of non-profit enterprises and several government research institutions. Will it be able to make another corporate Russian distribution? Probably, it will. But what about a stable and strong community? This question still remains open. At the moment, higher efficiency is shown by the middle-size private companies, which can create the finest products compared to those from state-owned enterprises. Miracle? Yes, it's true. Russia ignites technologies and specialists born to be professionals in Linux, but is unable to create one unified atmosphere for greater Linux government adoption—in schools, universities and within the government itself.

P.S. A small update: In September, the AltLinux company announced its withdrawal from the RASPO board. Either this decision was connected to fundamental differences between RASPO members, or the company decided to focus on Linux

Go, go, Linux!

Boston has adopted the best of Linux technologies. It is using Canonical's Ubuntu 12.10 and Red Hat's Fedora 17 development releases on Viridis, because these flavours apparently have very good support for Calxeda's version of ARM SoCs. The other ARM-based server products from Dell, HP and Mitac are also Linux-based.

Ganesan says, "Hardware is irrelevant without software. The data-centre companies are focused on leveraging open source software as broadly as possible, and so is the software strategy provided alongside Viridis. ARM and Calxeda have been working for multiple years to ready Linux distributions for the EnergyCore device."

"Linux was always our No 1 choice for OS, and it had to be that way," boasts Power! The majority of the WWW infrastructure is running some variant of a LAMP stack (<http://news.netcraft.com/archives/category/web-server-survey/>). Most of the world's clouds and cloud infrastructures are running Linux, with OpenStack being a good example. In the HPC space too, over 93 per cent of the world's most powerful supercomputers are also running Linux (<http://www.top500.org/statistics/list/>). Finally, the 'Big Data' space, which is dominated by Hadoop and Hadoop derivatives, is running Linux! With these being the main focus areas for Viridis, Linux was a de-facto choice! 

By: Janani Gopalakrishnan Vikram

The author is a technically-qualified freelance writer, editor and hands-on mom based in Chennai.

and not to disperse its efforts onto different initiatives—this is still unclear. Anyway, time will show who was right. And in August 2011, Red Hat opened its Moscow office. Although it doesn't yet plan to bring in development resources, no one can predict exactly how this will affect the corporate Linux sector in Russia. In general, we note just one thing: the current state of Linux in the local market reminds us of chaos. Amazingly, there's always something new that's born out of such chaos! 

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By: Anton Borisov

The author has specialised in Linux and FOSS technologies for more than a decade. His professional spheres of interest include, but are not limited to, robotics, embedded systems, statistics and algorithmic methods.